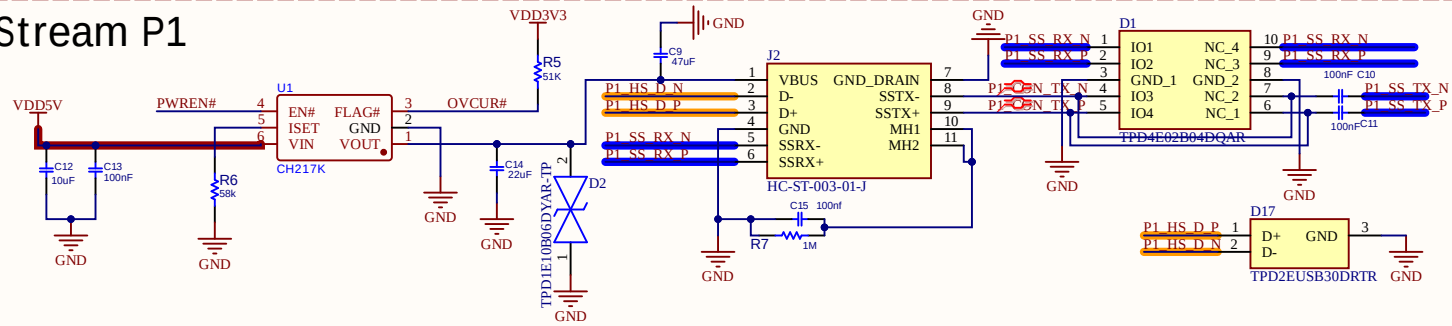
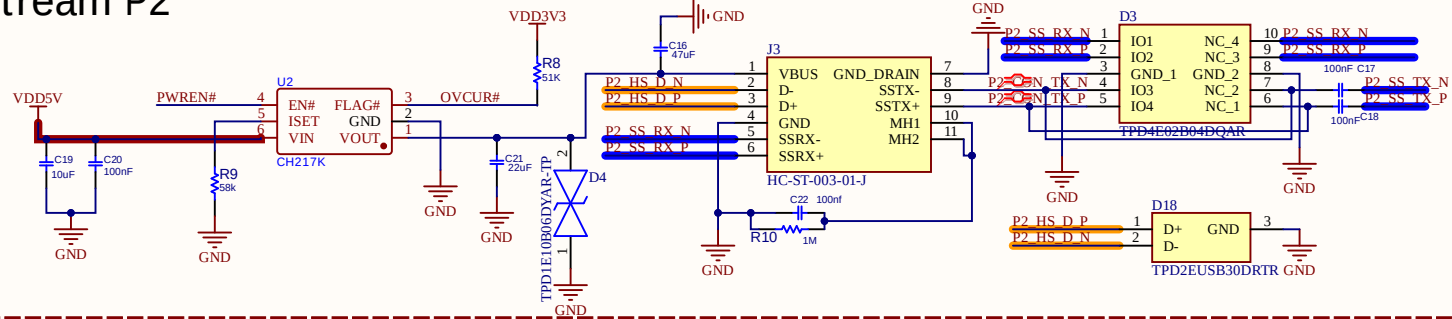


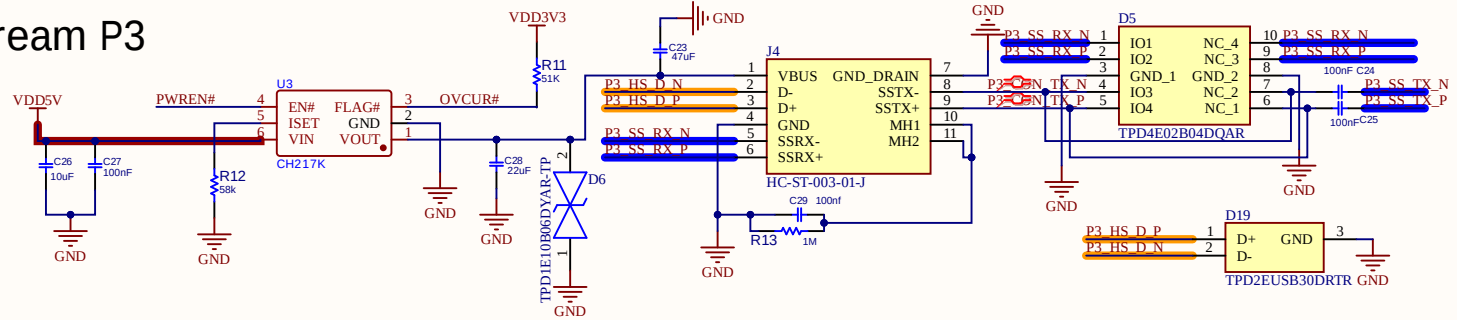
DownStream P1



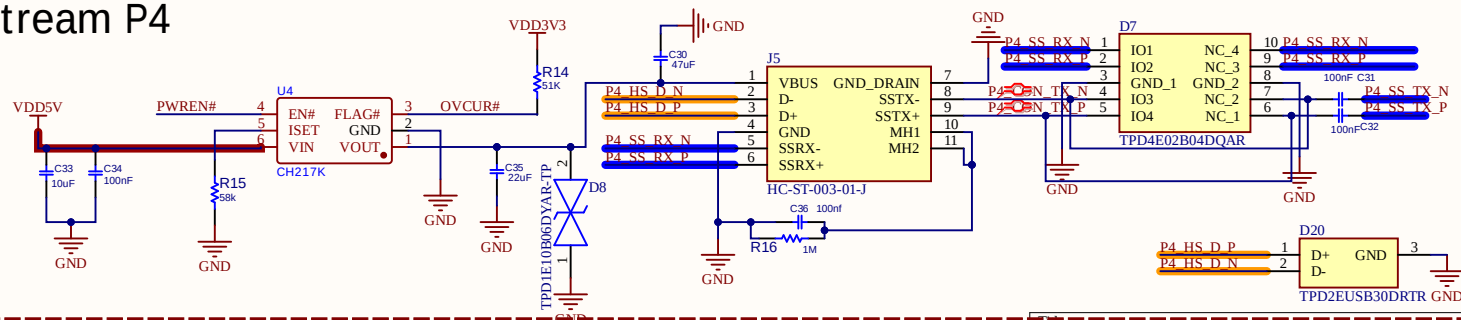
DownStream P2



DownStream P3

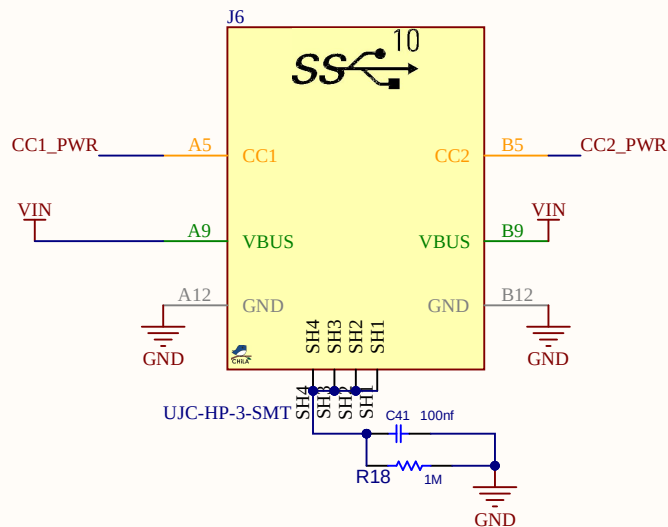


DownStream P4

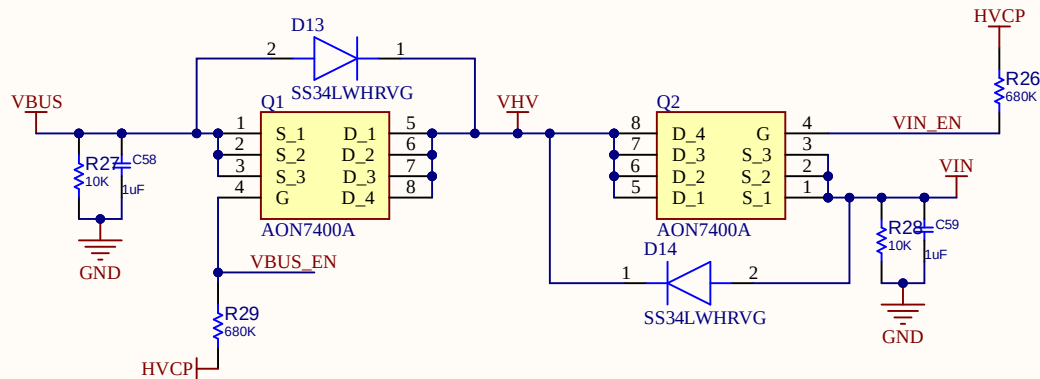


Size	Number	Revision
B		
Date:	7-06-2026	Sheet of
File:	D:\work\DownStream.SchDoc	Drawn By:

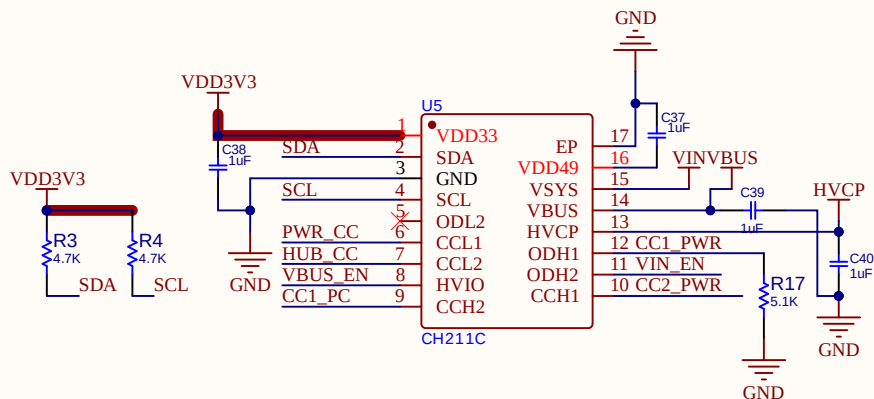
Type-C Power In



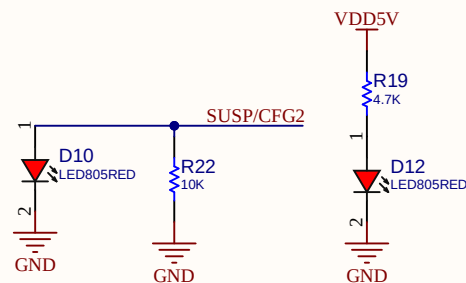
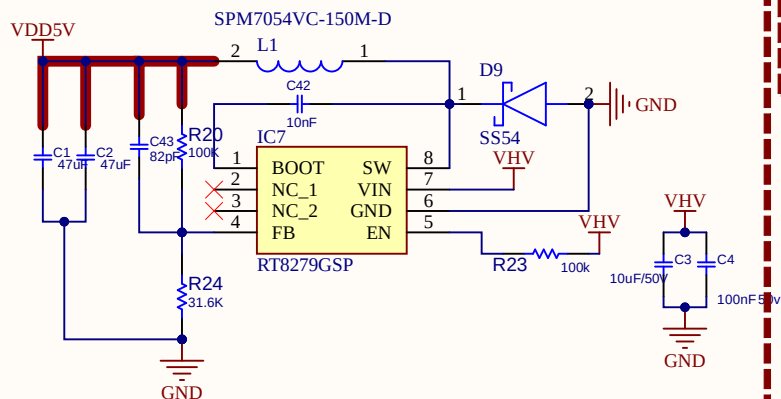
Power Control Switch



High Voltage Interface

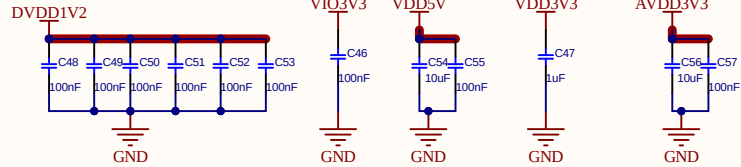


20V -> 5V Buck

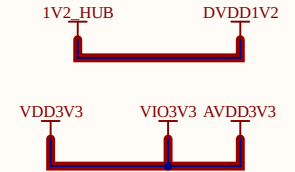
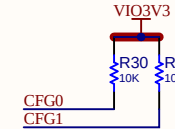
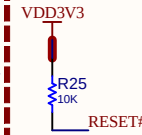
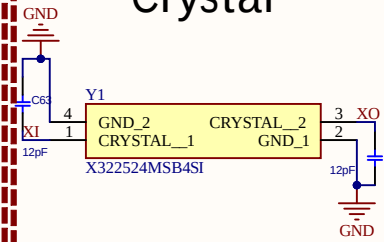


Title		
Size	Number	Revision
A		
Date:	7-06-2026	Sheet of
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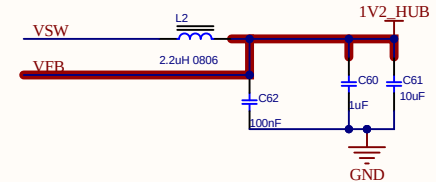
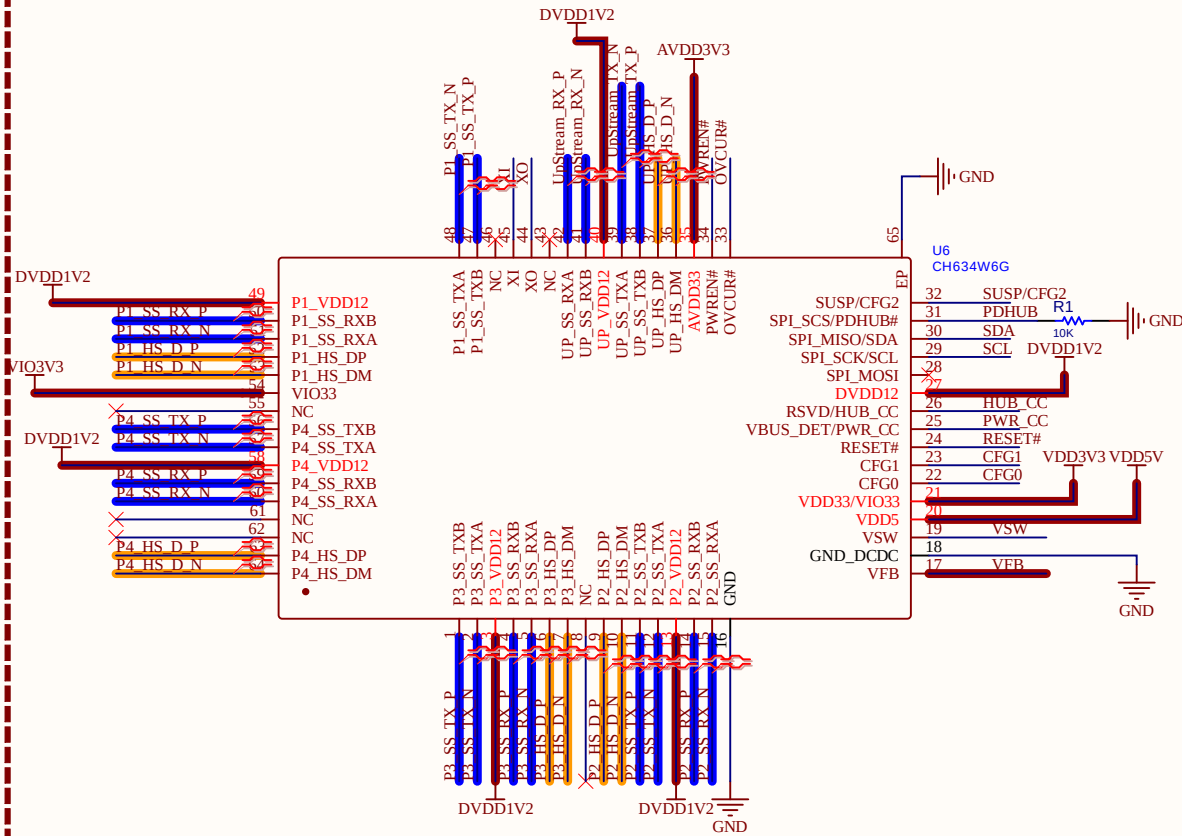
Decoupling Caps



Crystal



Interface Controller



Title		
Size	Number	Revision
A4		
Date:	7-06-2026	Sheet of
File:	D:\work\...\CH634W6G.SchDoc	Drawn By:

UP Stream Type-C Connector

SuperSpeed lines ESD

The diagram illustrates the ESD protection circuit for SuperSpeed lines using a D15 connector. The central component is the TPD4E02B04DQAR chip, which is connected to the connector pins as follows:

- Left Side (Pins 1-5):**
 - Pin 1: UpStream_TX_P
 - Pin 2: UpStream_TX_N
 - Pin 3: UpStream_RX_N
 - Pin 4: UpStream_RX_P (also connected to GND)
 - Pin 5: UpStream_RX_P
- Right Side (Pins 6-10):**
 - Pin 6: UpStream_RX_P
 - Pin 7: UpStream_RX_N (also connected to GND)
 - Pin 8: UpStream_TX_N
 - Pin 9: UpStream_TX_P
 - Pin 10: UpStream_TX_P

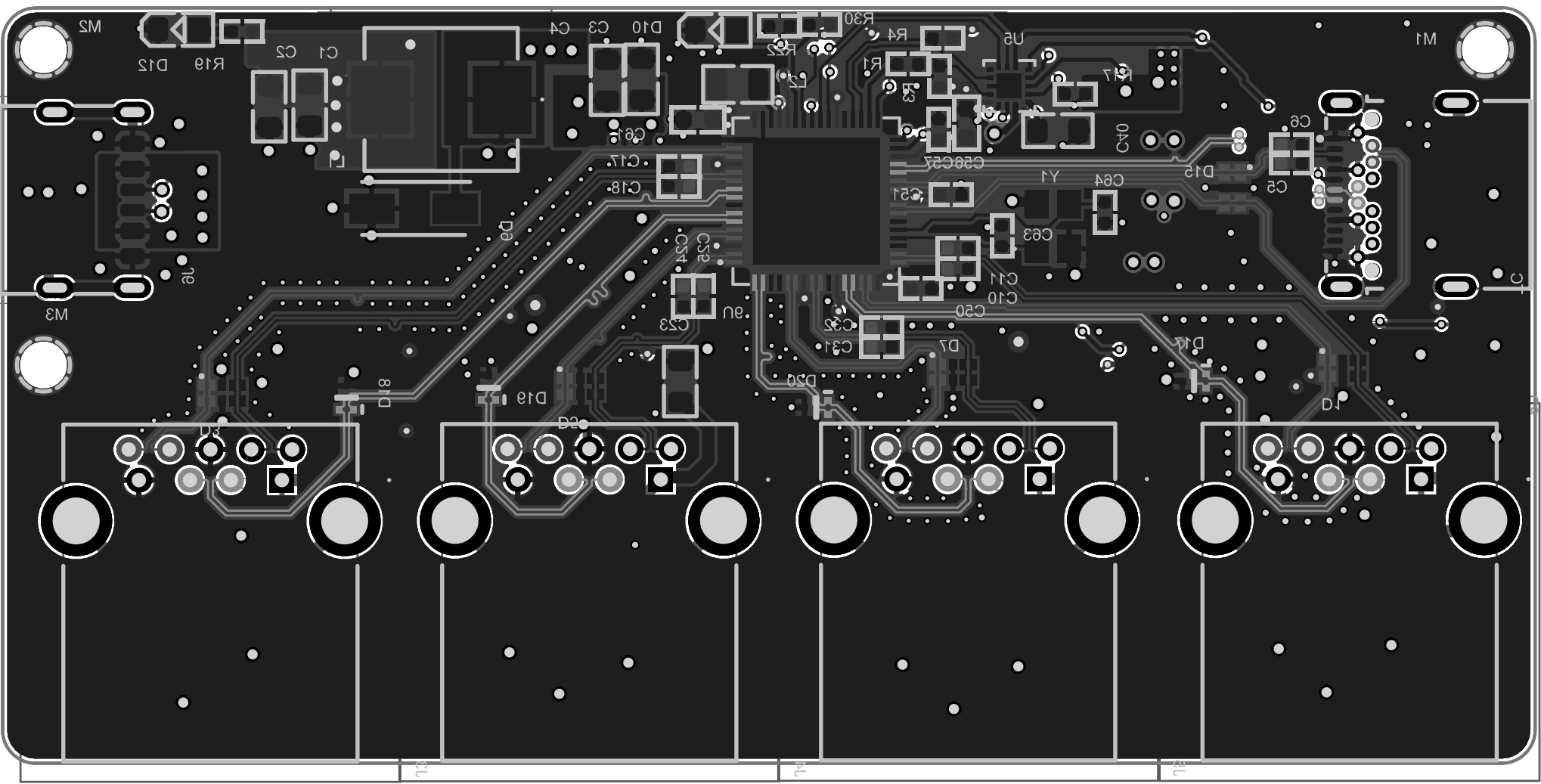
The chip is labeled **D15** and **TPD4E02B04DQAR**.

HighSpeed lines ESD

The diagram illustrates the ESD protection circuit for a USB Type-C connector. It features a yellow rectangular component labeled 'TPD2EUSB30DRTR' with three pins. Pin 1 is labeled 'D+' and Pin 2 is labeled 'D-'. Pin 3 is labeled 'GND'. The component is connected to a red ground symbol. The labels 'UP_HS_D_P' and 'UP_HS_D_N' are shown next to pins 1 and 2, respectively, with orange underlines. The label 'D21' is shown above the component. The label 'GND' is shown above the ground symbol.



Title		
Size A	Number	Revision
Date:	7-06-2026	Sheet of
File:	D:\work\...\UpStream.SchDoc	Drawn By:



Comment	Description	Designator	Footprint	LibRef	Quantity
47uF		C1, C2, C7, C9, C16, C23, C30	C0805	CAP_SMD_MLCC_0805	7
10uF/50V		C3	C0805	CAP_SMD_MLCC_0805	1
100nF50v	50V	C4	C0805	CAP_SMD_MLCC_0402	1
100nF		C5, C6, C10, C11, C13, C17, C18, C20, C24, C25, C27, C31, C32, C34, C46, C48, C49, C50, C51, C52, C53, C55, C57, C62	C0402	CAP_SMD_MLCC_0402	24
100nf	+1kv100nf	C8, C15, C22, C29, C36, C41	C1812	CAP_SMD_MLCC_1812	6
10uF	CAP, MLCC, -F, -V, MATERIAL, -%, 0201, [NoValue]	C12, C19, C26, C33, C54, C56, C61	C0603	CAP_SMD_MLCC_0603	7
22uF		C14, C21, C28, C35	C0805	CAP_SMD_MLCC_0805	4
1uF		C37, C38, C39, C40	C0805	CAP_SMD_MLCC_0805	4
10nF		C42	C0402	CAP_SMD_MLCC_0402	1
82pF		C43	C0402	CAP_SMD_MLCC_0402	1
1uF		C47, C58, C59, C60	C0402	CAP_SMD_MLCC_0402	4
12pF		C63, C64	C0402	CAP_SMD_MLCC_0402	2
TPD4E02B04DQAR	Diode	D1, D3, D5, D7, D15	DQA(R-PUSON-N10)	TPD4E02B04DQAR	5
TPD1E10B06DYAR-TP	TVS Diode (Bi-directional)	D2, D4, D6, D8, D16	SODFL1608X77N	TPD1E10B06DYARQ1	5
SS54	Schottky Diode	D9	DIOM5026X262N	SS54	1
LED0805RED	LED, SMD, 0805, RED	D10, D12	LED0805RED	LED_SMD_0805_RED	2
SS34LWHRVG	Diode	D13, D14	SODFL3718X102N	SS34LWHRVG	2
TPD2EUSB30DRTR	TPD2EUSB30ADRTR	D17, D18, D19, D20, D21	SOTFL35P100X50-3N	TPD2EUSB30DRTR	5
RT8279GSP	Integrated Circuit	IC7	SOIC127P600X175-8N	RT8279GSP	1
HC-ST-003-01-J	Connector	J2, J3, J4, J5	GSB4111312HR	GSB4111312HR	4
UIC-HP-3-SMT	CONN, USB, Type C (USB3.1), Right-Angle, SMD	J6	USBC_SMD_6_UICHP3_SMT	CONN_USBC_SMD_6_UIC-HP-3-SMT	1
SPM7054VC-150M-D	Inductor	L1	SPM7054VC150MD	SPM7054VC-150M-D	1
2.2uH 0806	SLW201610S2R2MST	L2	L0806	IND_SMD_1uH_2.3A_0806	1
AON7400A	MOSFET (N-Channel)	Q1, Q2	DFN_3X3_EP	AON7400A	2
10K	RESISTOR, SMD	R1, R22, R25, R27, R28, R30, R31	R0402	RES_SMD_0402	7
1M	RESISTOR, SMD, RESISTOR, SMD 0805	R2, R7, R10, R13, R16, R18	R0805	RES_SMD_0402	6
4.7K	RESISTOR, SMD	R3, R4, R19	R0402	RES_SMD_0402	3
51K	RESISTOR, SMD	R5, R8, R11, R14	R0402	RES_SMD_0402	4
58k	RESISTOR, SMD	R6, R9, R12, R15	R0402	RES_SMD_0402	4
5.1K	RESISTOR, SMD	R17	R0402	RES_SMD_0402	1
100K	RESISTOR, SMD	R20, R23	R0402	RES_SMD_0402	2
31.6K	RESISTOR, SMD	R24	R0402	RES_SMD_0402	1
680K	RESISTOR, SMD	R26, R29	R0402	RES_SMD_0402	2
10137065	CONN, USB, Type C, USB 3.1 Gen 1, Right-Angle, SMD	TC	USBC_HYB_U31CHG2SMT	CONN_USBC_SMD_10137065	1
CH217K		U1, U2, U3, U4	IC_IC-DP_SOT23-6L	CH217K	4
CH211C		U5	QFN-16C_L2.0-W2.0-P0.40-BL-EP1.2	CH211C	1
CH634W6G		U6	CH634w	CH634W6G	1
X322524MSB4S	Crystal or Oscillator	Y1	X1E0000210264	X1E0000210264	1